LEGO Engineering: Fall Term Lesson Plans

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Eight teach-tomorrow lessons for Term 1, built only on the LEGO Classic 10696 box and a cheap test kit.

How to use these plans

Every class runs the same 90 minutes plus recess. Each lesson below fills in this frame.

Block	Time	What happens
Morning Circle	15 min	3 min welcome and Superpower, 5 min Read & Learn (echo read, 3 words), 5 min demo, 2 min challenge intro
Build Challenge	25 min	Practice build at tables, using the challenge card
Recess	20 min	Bins closed, builds stay on the table
Competition Build	40 min	25 min build, 15 min testing and measuring in front of the class
Cleanup & Prizes	10 min	5 min sort, 5 min prizes, Superpower shout-out, 2 check questions

Challenge tiers. Every challenge card has three tiers. Everyone starts on Tier 1 (Beginner). A builder who finishes and passes the test moves to Tier 2, then Tier 3. Tiers follow skill, not grade, so a strong 2nd grader can reach Tier 3.

Grades 1 and 2. Read the passage to them and have them echo it. Builder Cards can be drawn and labeled. Give a 2-minute warning before every switch.

Grades 3 to 5. Students read aloud in popcorn style and write full Builder Card sentences. Give them the measuring jobs.

Seniors. Each lesson ends with a seniors note. Run it as 60 minutes in small teams of 2 or 3, with a real-world talking point. Skip prizes unless the group enjoys them. Memory care (45 min): one step at a time, larger bricks if you have them, and no timer.

Read & Learn. Use the existing Fall passages. They are under 150 words, so they fit the 5 minutes. Weeks 4 and 7 use the tablet-research and fable options from the literacy doc.

Test kit tote

One tote per instructor, reused every week at every site. Most of it is already in a house or a dollar store.

- 200 pennies in a zip bag
- 3 paper cups with string handles (punch 2 holes, tie 30 cm of string)
- 6 hardcover books of similar thickness
- Cloth tape measure and 2 rulers

- Painter's tape and a marker
- 2 pieces of stiff cardboard about 60 cm long (ramps)
- 1 cookie sheet or tray (shake table)
- Phone or stopwatch for timing
- Index cards (Builder Cards), sticky notes, pencils
- 4 file folders (privacy screens for Week 7)
- Small prize box (stickers, brick-shaped erasers, certificates)

Every-week setup (10 min before class): 1 bin of 10696 bricks per table of 3 to 4. Challenge cards printed and on tables. Vocabulary words on the board. Test station taped out at the front. Bins stay closed until the Build Challenge starts.

Week 1: Tower Power

Superpower: Organization. **Objective:** Students can explain that a wide base makes a tower more stable. **Vocabulary:** *stable* (stays standing without wobbling), *base* (the bottom part that holds everything up), *stud* (the bump on top of a brick that connects it). **Passage:** "What Is a LEGO Brick?" **Test kit:** tape measure, a book for the wobble test.

Morning Circle (15). Say: "Welcome, engineers! Engineers build things that work, and then they test them." Read the passage. Demo: hold up a tall skinny tower and a short wide one, and tap the table. "Which one fell? Why?" Let 2 or 3 kids answer, then say: "A wide base makes it *stable*."

Build Challenge (25). Sort first (5 min): each table sorts its bin by color into piles. Then build.

- Tier 1: a tower 20 cm tall that stands on its own.
- Tier 2: 30 cm tall, using 40 bricks or fewer.
- Tier 3: 30 cm tall, with a base no bigger than one 2x4 brick.

Competition Build (40). Tallest tower that passes the wobble test. Build for 25 min. To test, drop a book flat on the table 10 cm from the tower. If it stays up, measure it. Tallest standing tower wins. Tie-breaker: fewest bricks.

Check for understanding. "Why did the wide towers survive?" "Show me *stable* with your body."

When things go sideways. Towers keep falling: have the kid point to the base and ask how to make it wider. Early finishers: build a second tower and see whether they can beat their own height. Sorting gets messy: sort only reds and blues this week.

Seniors note. Teams of 3 build the tallest tower possible in 15 minutes. Talking point: "What's the tallest building you've been in? How do you think it stays up in the wind?"

Week 2: Strong Foundations

Superpower: Persistence. **Objective:** Students can show that overlapped bricks make a stronger wall and bridge than straight stacks. **Vocabulary:** *foundation* (the strong bottom layer a build sits on), *overlap* (a brick covers the seam between two bricks below), *load* (the weight a build has to hold). **Passage:** "How Builders Make Walls Strong." **Test kit:** 2 books per table, cup with string, 100+ pennies, ruler.

Morning Circle (15). Read the passage. Demo: two walls 6 bricks high, one stacked in straight columns and one overlapped like a brick wall. Ask: "Same bricks. Which one breaks first? Thumbs up for this one, thumbs down for that one." Push both. Say: "*Overlap* spreads the *load*."

Build Challenge (25).

- Tier 1: a wall 6 bricks high that survives a one-finger push.
- Tier 2: the same wall using 30 bricks or fewer.
- Tier 3: a wall with a doorway that still survives the push.

Competition Build (40). Penny bridge. Set two books 15 cm apart. Build a bridge across the gap, with nothing touching the table between the books. Hang the cup from the middle and add pennies 5 at a time. Most pennies before the bridge touches the table wins. Grades 1 and 2 count by 5s out loud as a class.

Check for understanding. "What does overlap do?" "Show me overlap with your hands."

When things go sideways. A bridge breaks at 5 pennies: give one rebuild, since that's the Persistence lesson. Early finishers: widen the gap to 20 cm. Short on long bricks: plates can be used for the deck.

Seniors note. Teams build bridges and compete on penny count. Talking point: local bridges people have crossed, and why old brick buildings still stand.

Week 3: Symmetry and Balance

Superpower: Attention to Detail. **Objective:** Students can explain that a symmetrical build balances because both sides weigh the same. **Vocabulary:** *symmetry* (both sides match like a mirror), *balance* (stays level without tipping), *center* (the middle point). **Passage:** "Symmetry All Around Us." **Test kit:** ruler, tape, small mirror if you have one.

Morning Circle (15). Read the passage. Demo: balance a ruler on one finger at its center. Then stick a 2x4 brick on one end and watch it tip. Ask: "How do I fix it?" (Put a matching brick on the other end.) Say: "*Symmetry* helps things *balance*."

Build Challenge (25). Mirror building with a partner. One partner builds half of a butterfly or robot face, and the other builds the mirror half.

- Tier 1: both halves use the same colors in the same spots.
- Tier 2: at least 15 bricks per side, all matching.
- Tier 3: a pattern that repeats 3 times on each side (for example red-blue-red).

Competition Build (40). Balancing bird. Build a creature with outstretched wings that balances on one 1x1 or 2x2 brick standing on the table. The longest balance wins (time it, and stop at 60 seconds). Score bonus points for wings that match exactly.

Check for understanding. "Why did the bird tip over when one wing was heavier?" "Point to the center of your build."

When things go sideways. Partners argue about who leads: flip a brick, and the partner whose color lands up builds first. Birds won't balance: move the support point toward the heavier side. Early finishers: make the wings longer.

Seniors note. Teams build balancing mobiles. Talking point: symmetry in quilts, architecture, and nature. People often share craft experience here.

Week 4: Creature Features

Superpower: Imagination. **Objective:** Students can name an animal feature and explain the job it does. **Vocabulary:** *feature* (a part with a special job, like a tail or claws), *habitat* (where an animal lives), *grip* (holding on without slipping). **Read & Learn:** Tablet research in pairs. Find 2 facts about one animal and write them on a sticky note. Grades 1 and 2: you read 3 short animal fact cards aloud and kids pick one. **Test kit:** cardboard ramp, 4 books, ruler.

Morning Circle (15). Do 5 minutes of research. Demo: show a picture of a mountain goat on a cliff. Ask: "How does it not fall?" (Wide hooves and a low body.) Say: "Every *feature* has a job."

Build Challenge (25). Build the animal you researched.

- Tier 1: it shows at least one real feature from the sticky note.
- Tier 2: it shows 2 features, and the Builder Card says what each one does.
- Tier 3: a made-up creature that borrows 3 features from real animals, each with a job.

Competition Build (40). Mountain goat climb. Build a creature that stands on a ramp without sliding or tipping over. Start the ramp on 1 book and add books one at a time. Grades 3 to 5 measure the height with a ruler. The creature still standing at the most books wins.

Check for understanding. "Name a feature and its job." "Which features helped your creature grip the ramp?"

When things go sideways. No tablets or no Wi-Fi: use the fact cards for everyone. Creatures slide right away: hint that a wider, lower body grips better. A kid wants to build only a dragon: fine, as long as it borrows a real feature.

Seniors note. Skip the tablets. Pair it with conversation about animals people have had or seen, and run the ramp test by teams. Memory care: build a simple animal together and talk about pets.

Week 5: Wheels and Axles

Superpower: Problem-Solving. **Objective:** Students can explain that wheels need to spin freely on an axle for a car to roll far. **Vocabulary:** *axle* (the rod or piece a wheel spins on), *wheel* (a round part that turns to move a load), *ramp* (a slope that gives a car speed). **Passage:** "Wheels That Changed the World." **Test kit:** 2 cardboard ramps on 2 books each, painter's tape start line, tape measure.

Morning Circle (15). Read the passage. Demo: roll two cars down the ramp. One has a wheel rubbing against the body. Ask: "Why did this one stop?" Say: "Wheels need room to spin on the *axle*."

Build Challenge (25). Use the wheels in the 10696 box. If a site is short, lend from your wheel bin this week only.

- Tier 1: a car with 4 wheels that rolls straight off the ramp.
- Tier 2: a car that carries one minifigure or a 2x2 "passenger" without losing it.
- Tier 3: a car that rolls at least 1 meter past the end of the ramp.

Competition Build (40). Ramp race for distance. Each car gets 2 runs from the same start line, and you measure from the end of the ramp to where the front of the car stops. Best run counts. Bonus round: add a 10-penny load in a cup on top and race again.

Check for understanding. "What made the farthest car fast?" "Point to the axle on your car."

When things go sideways. A car veers: check whether the wheels are lined up. Not enough wheels: build in pairs. Kids crowd the ramp: tape a waiting line, and only the racer stands at the ramp.

Seniors note. Teams race. Talking point: first cars people drove, and how roads and cars have changed. Very social and competitive in a friendly way.

Week 6: Architects and Earthquakes

Superpower: Planning. **Objective:** Students can sketch a plan before building and explain one way to make a building earthquake-safe. **Vocabulary:** *architect* (a person who designs buildings), *blueprint* (a drawing that shows the plan), *earthquake* (when the ground shakes). **Passage:** "What Does an Architect Do?" **Test kit:** cookie sheet (shake table), timer, half sheets of paper for sketches.

Morning Circle (15). Read the passage. Demo: put a tall thin house and a short wide house on the cookie sheet and slide it back and forth for 5 seconds. Ask: "Which one would you live in?" Say: "*Architects* plan first. That's why we draw a *blueprint*."

Build Challenge (25). First 5 minutes: sketch a house with a door, windows, and a roof. Then build it from the sketch.

- Tier 1: the house matches the sketch (door, 2 windows, roof).
- Tier 2: a 2-story house with a floor between the stories.
- Tier 3: an interior room with furniture, and a roof that lifts off to show it.

Competition Build (40). Shake table. Build the tallest house that survives 10 seconds of shaking. You do the shaking each time, at the same speed and distance (count "1 shake per second, 3 cm each way"). Survivors get measured, and the tallest wins. At the end, connect all the houses into one neighborhood on the floor.

Check for understanding. "Why did we draw before building?" "What helped the winning house survive the shake?"

When things go sideways. Kids skip the sketch: no bricks until you've seen the paper. The tray is too wild: shake gentler, but shake the same for everyone. A crying kid when their house falls: point to the Superpower and give them 2 minutes to rebuild.

Seniors note. Teams design a dream home or the house they grew up in. Talking point: homes people have lived in. Run the shake test only if the group wants it.

Week 7: Story Builders

Superpower: Communication. **Objective:** Students can give clear building directions using position words (on top, next to, under). **Vocabulary:** *character* (who the story is about), *setting* (where and when it happens), *instructions* (steps that tell someone exactly what to do). **Read & Learn:** Read a short fable aloud together, such as "The Tortoise and the Hare." **Test kit:** file folders as privacy screens, index cards.

Morning Circle (15). Read the fable. Demo: hide a 4-brick model behind a folder and describe it badly ("put a brick on a brick"), and have a volunteer try to copy it. Then describe it well ("a red 2x4 on the bottom, a blue 2x2 on top, on the left end"). Say: "Engineers write *instructions* so anyone can build their idea."

Build Challenge (25). Build a scene from the fable or your own story, with a character and a setting. Then write a 2-sentence caption on the Builder Card.

- Tier 1: the scene shows one character and a setting.
- Tier 2: the scene shows the problem in the story.
- Tier 3: two scenes, the problem and how it was solved.

Competition Build (40). Blind build. Partners sit back to back, or with a folder between them. The Builder secretly makes a 6-brick model (Tier 2: 8 bricks, Tier 3: 10). The Describer can't touch or show, only talk. The partner builds from the words. Score 1 point for each brick in the right spot. Then swap roles. Highest pair total wins. Before cleanup, 3 volunteers read their story captions aloud.

Check for understanding. "What words helped your partner most?" "What's the setting of your scene?"

When things go sideways. Grades 1 and 2 find 6 bricks too hard: start with 3, all different colors. Peeking: the folder stays up and you referee. Odd number of kids: you be a partner.

Seniors note. Blind build is a great fit for seniors: it's social, funny, and good for memory and language. Talking point: a favorite story from childhood. Memory care: use 3 large bricks and describe colors only.

Week 8: Engineering Showcase

Superpower: Pride. **Objective:** Students can present one build and explain the engineering idea behind it using a vocabulary word from the term. **Vocabulary:** Review all 21 words. New word: *engineer* (a person who designs, builds, tests, and improves). **Read & Learn:** Each student reads their favorite past Builder Card to a partner. **Test kit:** everything from the term, set up as stations.

Morning Circle (15). Walk through the 7 Superpowers. Quick vocabulary game: you say a definition, and kids shout the word. Explain the Showcase: "Today you pick your favorite test from the term and try to beat your own record."

Build Challenge (25). Rematch. Each student picks one past challenge (tower, penny bridge, balancing bird, ramp car, or shake house) and tries to beat their own score. Tier 3 builders can help a younger or newer builder after they finish.

Competition Build (40). Showcase and gallery. 20 min: build a personal masterpiece with a theme. 20 min: gallery walk. Each builder stands by their build and reads their Builder Card, which says "My build is ___ because ___" with one vocabulary word. If families attend, they walk the gallery too.

Cleanup & Prizes (10). A certificate for everyone, naming the Superpower they showed most. Awards: Most Improved Record, Best Engineer Explanation, Persistence Award.

Check for understanding. Each student says one word they learned this term and what it means.

When things go sideways. Absent kids missed a challenge: let them pick any challenge they did attend. Families arrive early: hand them a scorecard and invite them to vote. Running long: skip the masterpiece and use the rematch builds for the gallery.

Seniors note. Hold a showcase for residents and staff. Each team explains its favorite build. A good moment to invite families or post photos in the community newsletter, with consent.